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On General Adaptive Sparse Principal Component Analysis

Chenlei LENG and Hansheng WANG

The method of *sparse principal component analysis* (S-PCA) proposed by Zou, Hastie, and Tibshirani (2006) is an attractive approach to obtain sparse loadings in principal component analysis (PCA). S-PCA was motivated by reformulating PCA as a least-squares problem so that a lasso penalty on the loading coefficients can be applied. In this article, we propose new estimates to improve S-PCA in the following two aspects. First, we propose a method of *simple adaptive sparse principal component analysis* (SAS-PCA), which uses the adaptive lasso penalty (Zou 2006; Wang, Li, and Jiang 2007) instead of the lasso penalty in S-PCA. Second, we replace the least-squares objective function in S-PCA by a general least-squares objective function. This formulation allows us to study many related sparse PCA estimators under a unified theoretical framework and leads to the method of *general adaptive sparse principal component analysis* (GAS-PCA). Compared with SAS-PCA, GAS-PCA enjoys much improved finite sample performance. In addition, we show that, when a BIC-type criterion is used for selecting the tuning parameters, the resulting estimates are consistent in variable selection. Numerical studies are conducted to compare the finite sample performance of various competing methods.

Datasets and computer code are available in the online supplements.

Key Words: Adaptive lasso; BIC; GAS-PCA; LARS; Lasso; SAS-PCA; S-PCA.

1. INTRODUCTION

Principal component analysis (PCA) is an important quantitative tool for studying the variance-covariance structure of multivariate responses, with successful applications in many different fields (Johnson and Wichern 2003).

For linear regression analysis, estimation efficiency and model parsimony are the two major concerns in fitting a meaningful model (Yang 2005). Nevertheless, as for PCA, parsimony is particularly relevant because an easily interpretable result can help the researcher to not only decide the number of the important components but also identify interesting

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loading patterns. From there one can infer which set of variables are highly correlated with which component, and thereafter, a sensible *factor* might be extracted.

Typically, the number of important principal components can be determined by studying their contribution to explaining the total variability (Johnson and Wichern 2003). Nevertheless, how to decide significant loading coefficients is much more difficult because the ordinary PCA cannot produce sparse solutions. In addition, no simple yet adequate testing procedure exists for those estimated loading coefficients. Hence, the interpretation of the corresponding loading patterns is difficult. On the other hand, for better interpretation, one has to decide which variable is heavily loaded on which component. To this end, some heuristic thresholding criteria are used, which are highly subjective and not generally applicable.

To address this question, Zou, Hastie, and Tibshirani (2006) proposed a shrinkage estimator for the principal component analysis by reformulating it as an equivalent least-squares problem. In this formulation, the classical *least absolute shrinkage and selection operator* (lasso) can be directly applied (Tibshirani 1996) and sparse loadings can be obtained. We refer to methods of this type as *sparse principal component analysis* (S-PCA). The goal of this article is to make several improvements to this important method. First, we replace the lasso penalty utilized in S-PCA by the adaptive lasso penalty (Zou, 2006; Wang, Li, and Jiang 2007), which leads to the method of *simple adaptive sparse principal component analysis* (SAS-PCA). Compared with S-PCA, SAS-PCA uses adaptive shrinkage for different loading coefficients that takes account of their estimated relative importance, which naturally improves both estimation efficiency and selection accuracy. Second, we replace the least-squares objective function in S-PCA by a general least-squares objective function. This formulation allows us to study many related sparse PCA estimators under a unified theoretical framework and leads to the method of *general adaptive sparse principal component analysis* (GAS-PCA). Compared with SAS-PCA, GAS-PCA enjoys much improved finite sample performance. In addition, we show that when a BIC-type criterion is used for selecting the tuning parameters, the resulting estimator is consistent in variable selection.

The rest of the article is organized as follows. The next section introduces the model notation, the SAS-PCA method, and the GAS-PCA method. Their main theoretical properties are discussed in Section 3. Numerical studies are provided in Section 4 and the article concludes with a short discussion in Section 5. The program and the dataset used for this work can be obtained from the JCGS official website.

2. METHODOLOGY DEVELOPMENT

2.1 THE S-PCA METHOD

Let $\mathbf{X} = (X_1, \ldots, X_d)^\top \in \mathcal{R}^d$ be a d-dimensional random vector and $\mathbf{x}_i = (x_{i1}, \ldots, x_{id})^\top \in \mathcal{R}^d$ be its realization on the ith subject ($i = 1, \ldots, n$). Without loss of generality we assume that $\mathrm{E}(\mathbf{X}) = 0$ and $\mathrm{cov}(\mathbf{X}) = \Sigma$ for some positive definite matrix Σ, which can be represented as

$$\Sigma = \delta_1 \beta_1 \beta_1^\top + \cdots + \delta_d \beta_d \beta_d^\top, \tag{2.1}$$

where δ_j is the jth largest eigenvalue of Σ and $\beta_j = (\beta_{j1}, \ldots, \beta_{jd})^\top \in \mathcal{R}^d$ is its associated eigenvector. For the purpose of identifiability, we assume that $\delta_1 > \delta_2 > \cdots > \delta_d > 0$ and that the first nonzero component of β_j is positive. Under this condition, $\{\beta_j\}_{j=1}^d$ are uniquely determined by Σ and are orthonormal. The jth principal component of $\mathbf{x}_i$ is given by $\mathbf{x}_i^\top \beta_j$, which accounts for $100 \times \delta_j / (\sum_{j'} \delta_{j'})$ percent of the total variance $\sum_j \mathrm{var}(X_j)$.

In data analysis, we can estimate Σ and β_j by their sample versions

$$\tilde{\Sigma} = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i^\top = \sum_{j=1}^d \tilde{\delta}_j \tilde{\beta}_j \tilde{\beta}_j^\top,$$

where $\tilde{\delta}_j$ and $\tilde{\beta}_j = (\tilde{\beta}_{j1}, \ldots, \tilde{\beta}_{jd})^\top \in \mathcal{R}^d$ stand for the eigenvalue–eigenvector decomposition of $\tilde{\Sigma}$. For convenience, we refer to $(\beta_1, \ldots, \beta_d) \in \mathcal{R}^{d \times d}$ as the true loading coefficients and $(\tilde{\beta}_1, \ldots, \tilde{\beta}_d) \in \mathcal{R}^{d \times d}$ as its unpenalized sample estimates. In applications, usually it is theoretically yet practically attractive to assume that the true loading coefficients of the first few (e.g., $d_0 \ll d$) leading principal components are sparse. In particular, for each $\beta_j = (\beta_{j1}, \ldots, \beta_{jd})^\top$, an appropriate sparse representation usually implies the existence of a meaningful latent variable (i.e., *factor*), with which the responses having nonzero loadings are highly correlated. If such a loading pattern can be easily interpreted, a practically meaningful *factor* might be extracted.

Nevertheless, the standard principal component estimate $\tilde{\beta}_j$ cannot produce sparse solutions. As a novel solution, Zou, Hastie, and Tibshirani (2006) developed the *sparse principal component analysis* (S-PCA) methodology. Specifically, by focusing on the first d_0 ($d_0 < d$) leading principal components, they transformed the original problem into the following equivalent least-squares problem:

$$(\hat{A}, \hat{B}) = \mathrm{argmin} \left\{ \frac{1}{n} \sum_{i=1}^n \|\mathbf{x}_i - A B^\top \mathbf{x}_i\|^2 \right\} \quad \text{subject to } A^\top A = I, \tag{2.2}$$

where $A = (\alpha_1, \ldots, \alpha_{d_0}) \in \mathcal{R}^{d \times d_0}$, $B = (\beta_1, \ldots, \beta_{d_0}) \in \mathcal{R}^{d \times d_0}$, and $I \in \mathcal{R}^{d_0 \times d_0}$ is the identity matrix. Let the corresponding estimates be $\hat{A} = (\hat{\alpha}_1, \ldots, \hat{\alpha}_{d_0})$ and $\hat{B} = (\hat{\beta}_1, \ldots, \hat{\beta}_{d_0})$, respectively. The formulation in (2.2) motivates the lasso method (Tibshirani 1996) as follows:

$$(\hat{A}_{\bar{\lambda}}, \hat{B}_{\bar{\lambda}}) = \mathrm{argmin} \left\{ \frac{1}{n} \sum_{i=1}^n \|\mathbf{x}_i - A B^\top \mathbf{x}_i\|^2 + \sum_{j=1}^{d_0} \left(\lambda_j \sum_{k=1}^d |\beta_{jk}| \right) \right\}$$
$$\text{subject to } A^\top A = I, \tag{2.3}$$

where $\bar{\lambda} = (\lambda_1, \ldots, \lambda_{d_0})^\top$, $\|\cdot\|$ stands for the usual L_2 norm, and $\hat{A}_{\bar{\lambda}} = (\hat{\alpha}_{\bar{\lambda},1}, \ldots, \hat{\alpha}_{\bar{\lambda},d_0})$, $\hat{B}_{\bar{\lambda}} = (\hat{\beta}_{\bar{\lambda},1}, \ldots, \hat{\beta}_{\bar{\lambda},d_0})$ are the resulting lasso estimator. We refer to this approach as *sparse principal component analysis* (S-PCA) and the resulting estimator $\hat{B}_{\bar{\lambda}}$ as the S-PCA estimator.

2.2 THE SAS-PCA METHOD

The S-PCA method of Zou, Hastie, and Tibshirani (2006) may not be fully adaptive. For the situation that the dimension of the response is substantially smaller than the number of observations (i.e., $d \ll n$), the excessive shrinkage equally applied by lasso to each coefficient seems to be problematic, at least in the least-squares setting (Fan and Li 2001; Yuan and Lin 2005; Leng, Lin, and Wahba 2006; Zou, 2006). A simple solution, proposed by Zou (2006) and Wang, Li, and Tsai (2007a), is to modify the lasso penalty so that different shrinkage can be used for different coefficients, leading to a consistent selection of the important coefficients with high efficiency. Thus, this method is referred to as the adaptive lasso by Zou (2006). Moreover, no automatic yet objective tuning parameter selector is available for S-PCA. As an alternative, Zou, Hastie, and Tibshirani (2006) relied on inspection by eye to balance the goodness of fit and the sparsity level.

The success of the adaptive lasso in regression motivates the following adaptive lasso method:

$$(\hat{A}_\lambda, \hat{B}_\lambda) = \operatorname{argmin}\left\{\frac{1}{n}\sum_{i=1}^{n}\|\mathbf{x}_i - AB^\top \mathbf{x}_i\|^2 + \sum_{j=1}^{d_0}\sum_{k=1}^{d}\lambda_{jk}|\beta_{jk}|\right\}$$
$$\text{subject to } A^\top A = I, \tag{2.4}$$

where $\lambda = \{\lambda_{jk} : 1 \le j \le d_0; 1 \le k \le d\} \in \mathcal{R}^{d_0 \times d}$. Comparing (2.4) and (2.3), the key difference is that (2.4) allows different tuning parameters (hence adaptive shrinkage) for different loading coefficients. Intuitively, if more shrinkage is used for the zero coefficients with less shrinkage for the nonzero ones, an estimator with higher efficiency may be obtained. One can refer to Zou (2006) for a more detailed and rigorous discussion about the advantages of the adaptive lasso penalty over the traditional one. We refer to this approach as the *simple adaptive sparse principal component analysis* (SAS-PCA), and refer to $\hat{B}_\lambda = (\hat{\beta}_{\lambda 1}, \ldots, \hat{\beta}_{\lambda d_0})$ as the SAS-PCA estimator.

Following Zou, Hastie, and Tibshirani (2006), the SAS-PCA estimator can be obtained by optimizing the objective function (2.4) with respect to A and B iteratively. Specifically, for a fixed B, the value of A is updated as $A = UV^\top$ where $\tilde{\Sigma}B = UDV^\top$ is the singular value decomposition of $\tilde{\Sigma}B$. More details can be found in theorem 4 of Zou, Hastie, and Tibshirani (2006). Now, for a fixed A, simple linear algebra shows that, up to a constant irrelevant to B, the objective function (2.4) is equivalent to

$$\sum_{j=1}^{d_0}\left\{\frac{1}{n}\sum_{i=1}^{n}(\alpha_j^\top \mathbf{x}_i - \beta_j^\top \mathbf{x}_i)^2 + \sum_{k=1}^{d}\lambda_{jk}|\beta_{jk}|\right\}$$
$$= \sum_{j=1}^{d_0}\left\{(\alpha_j - \beta_j)^\top \tilde{\Sigma}(\alpha_j - \beta_j) + \sum_{k=1}^{d}\lambda_{jk}|\beta_{jk}|\right\}, \tag{2.5}$$

which consists of d_0 independent adaptive lasso problems (Zou, 2006). For each fixed j, the entire solution path can be easily obtained by the fast LARS algorithm (Efron et al. 2004), from which the optimal solution can be easily identified according to the following

BIC-type selection criterion (Wang and Leng 2007; Wang, Li, and Tsai 2007b):

$$\mathrm{BIC}_{\lambda j} = (\hat{\alpha}_{\lambda j} - \hat{\beta}_{\lambda j})^\top \tilde{\Sigma} (\hat{\alpha}_{\lambda j} - \hat{\beta}_{\lambda_j}) + df_{\lambda j} \times \frac{\log n}{n}. \tag{2.6}$$

Here df_{λ_j} is the number of nonzero coefficients identified in $\hat{\beta}_{\lambda j}$ (Zou, Hastie, and Tibshirani 2007). In fact, by following the arguments in Wang and Leng (2007), we shall show in the next section that the tuning parameters selected by BIC can identify all sparse coefficients consistently.

In practice, simultaneously tuning so many regularization parameters is computationally expensive and practically infeasible. Hence, we apply the idea in Zou (2006) and Wang, Li, and Tsai (2007a), and propose the following simple estimate $\lambda_{jk} = \tau_j / |\tilde{\beta}_{jk}|$ for some $\tau_j > 0$. Recall that $\tilde{\beta}_{jk}$ is the loading coefficient obtained by the ordinary principal component analysis. Hence, the original $d_0 \times d$ regularization parameters reduce to only d_0 choices (i.e., $\{\tau_j, 1 \le j \le d_0\}$), which can be decided separately according to (2.6). Here we implicitly assume that different regularization parameters, τ_j, can be used for different eigenvectors β_j. Thus, we refer to this tuning parameter selection scheme as the local BIC method. For simplicity, one can also force $\tau_j = \tau_0$ for some $\tau_0 > 0$. The original problem of selecting $d_0 \times d$ tuning parameters then reduces to one univariate optimization problem for τ_0. We refer to this selection scheme as the global BIC method. Although the asymptotic properties of the global BIC and local BIC are similar, as discussed in the next section, we find that the local BIC method typically performs better than the global BIC method (see Example 3 in Section 4). The better performance of local BIC is not surprising because it is more adaptive to the different sparsity levels of the eigenvectors.

2.3 THE GAS-PCA METHOD

Our numerical experience suggests that SAS-PCA performs well if the zeros and nonzero coefficients are well separated. However, if this is not the case, the finite sample performance of SAS-PCA may suffer (see Example 1 in Section 4), although SAS-PCA usually outperforms S-PCA. This observation confirms the conjecture of an anonymous referee and motivates us to further improve SAS-PCA, which leads to the following method of *general adaptive sparse principal component analysis* (GAS-PCA).

Simply speaking, the formulations of both (2.5) and (2.6) suggest a general class of methods for sparse principal component analysis. More specifically, given the matrix A, instead of updating B by minimizing the objective function (2.5), we can minimize the following general least-squares objective function:

$$\sum_{j=1}^{d_0} \left\{ (\alpha_j - \beta_j)^\top \tilde{\Omega} (\alpha_j - \beta_j) + \sum_{k=1}^{d} \lambda_{jk} |\beta_{jk}| \right\}, \tag{2.7}$$

where $\tilde{\Omega}$ is a positive definite matrix with a probabilistic limit Ω, a positive definite matrix referred to as the kernel matrix. Once β_j is estimated, α_j can be updated in a manner similar to that in Zou, Hastie, and Tibshirani (2006). One can iterate between those two steps until convergence occurs. The BIC criterion associated with this method is given by

$$\mathrm{BIC}_{\lambda j} = (\hat{\alpha}_{\lambda j} - \hat{\beta}_{\lambda j})^\top \tilde{\Omega} (\hat{\alpha}_{\lambda j} - \hat{\beta}_{\lambda_j}) + df_{\lambda j} \times \frac{\log n}{n}. \tag{2.8}$$

We remark that if we fix $\tilde{\Omega} = \tilde{\Sigma}$, then (2.7) and (2.8) reduce to (2.5) and (2.6), respectively. Thus, the new formulations in (2.7) and (2.8) are more general and include both S-PCA and SAS-PCA as special cases. Our numerical experience suggests that the choice of the matrix $\tilde{\Omega}$ plays an important role in the finite sample performance, and that $\tilde{\Omega} = \tilde{\Sigma}$ (i.e., the SAS-PCA method) is unlikely the optimal choice. This immediately raises a question: what $\tilde{\Omega}$ should be used in (2.7)?

Recently, Wang and Leng (2007) proposed the method of least-squares approximation (LSA), which gives adaptive lasso estimators for a wide class of regression models. Simply speaking, Wang and Leng (2007) found that the estimator produced by minimizing the following least-squares–type objective function:

$$(\hat{\theta} - \theta)^{\top} \{\widehat{\mathrm{cov}}^{-1}(\hat{\theta})\}(\hat{\theta} - \theta) + \sum_{k=1}^{d} \lambda_k |\theta_k| \tag{2.9}$$

possesses excellent finite sample properties and well-established asymptotic theory, where $\theta = (\theta_1, \ldots, \theta_d)^{\top} \in \mathcal{R}^d$ is some unknown parameter, $\hat{\theta}$ is its unpenalized estimator (e.g., maximum likelihood estimator), and $\widehat{\mathrm{cov}}(\hat{\theta})$ is an estimator of $\mathrm{cov}(\hat{\theta})$. The GAS-PCA formulation in (2.7) is similar to the LSA formulation in (2.9). In particular, if we replace the unpenalized estimator $\hat{\theta}$ in (2.9) by the unpenalized eigenvector estimate $\tilde{\beta}_j$, we can replace $\tilde{\Omega}$ by $\mathrm{cov}^{-1}(\tilde{\beta}_j)$. Unfortunately, no simple formula exists for $\mathrm{cov}(\tilde{\beta}_j)$, so we propose the following bootstrap method to estimate $\mathrm{cov}(\tilde{\beta}_j)$. For given $\tilde{\Sigma}$, we apply PCA to bootstrap samples from $N(0, \tilde{\Sigma})$ to produce a sufficient number of bootstrap estimates (denoted by $\hat{\beta}_j^{boot}$) for β_j. The sample covariance of $\hat{\beta}_j^{boot}$ can be easily computed (denoted by $\hat{C}_j$) as a natural estimate of $\mathrm{cov}(\tilde{\beta}_j)$. We then follow Wang and Leng (2007) and fix $\tilde{\Omega} = \hat{C}_j^{-1}$. Thereafter, the objective function (2.7) can be optimized iteratively in the same manner as (2.5). This approach is referred to as *general adaptive sparse principal component analysis* (GAS-PCA). For simplicity, we denote the resulting estimator by $\hat{B}_{\lambda}^{*} = (\hat{\beta}_{\lambda j}^{*}, \ldots, \hat{\beta}_{\lambda d_0}^{*})$ and refer to it as the GAS-PCA estimator. Our theoretical analysis (see Theorem 3 in Section 3.2) and numerical experiments (see examples in Section 4) suggest that the GAS-PCA estimator further improves upon the SAS-PCA estimator in terms of estimation efficiency and selection consistency, sometimes substantially.

Last, motivated by an anonymous referee's comment, we also study the relationship between the new formulation in (2.7) and soft-thresholding methods. It turns out that soft-thresholding rules are also special cases of (2.7). For example, if we ignore the covariance structure by taking $\tilde{\Omega}$ to be a diagonal matrix such as $\tilde{\Omega} = \mathrm{diag}\{\widehat{\mathrm{cov}}^{-1}(\hat{\theta})\} = \mathrm{diag}\{\hat{\sigma}_1^2, \ldots, \hat{\sigma}_d^2\}$, then (2.7) becomes

$$\sum_{j=1}^{d_0} \left\{ \sum_{k=1}^{d} \left(\frac{(\alpha_{jk} - \beta_{jk})^2}{\hat{\sigma}_{jk}^2} + \lambda_{jk} |\beta_{jk}| \right) \right\}, \tag{2.10}$$

which leads to the following simple thresholding rule (Tibshirani 1996): $\beta_{jk} = \mathrm{sign}(\alpha_{jk}) \times [|\alpha_{jk}| - 0.5\lambda_{jk}\hat{\sigma}_{jk}^2]_+$, where $[t]_+ = tI(t > 0)$. We refer to this estimator as the *variance adjusted soft threshold* (VAST) estimator. As one can see, (2.7) allows us to formulate S-PCA, SAS-PCA, GAS-PCA, and VAST under a single unified theoretical framework, which greatly facilitates the theoretical treatment of these various methods.

3. THEORETICAL PROPERTIES

3.1 BASIC THEORETICAL PROPERTIES

Because S-PCA, SAS-PCA, GAS-PCA, and VAST are special cases of (2.7), we shall discuss the asymptotic properties of a general estimator given by minimizing (2.7). Define

$$a_n = \max\{\lambda_{jk} : \beta_{jk} \neq 0 : 1 \leq j \leq d_0, 1 \leq k \leq d\},$$
$$b_n = \min\{\lambda_{jk} : \beta_{jk} = 0 : 1 \leq j \leq d_0, 1 \leq k \leq d\}.$$

Thus a_n is the maximum shrinkage applied to the significant coefficients and b_n is the minimum shrinkage for the insignificant ones. To understand the proposed methods, we focus on the situation where $\hat{\alpha}_{\lambda j}$ is fixed to be $\bar{\alpha}_j$ for some $\bar{\alpha}_j \in \mathcal{R}^d$. Define

$$\bar{\beta}_{\lambda j} = (\bar{\beta}_{\lambda jk}) = \underset{\beta_j}{\operatorname{argmin}} \left\{ (\bar{\alpha}_j - \beta_j)^\top \tilde{\Omega} (\bar{\alpha}_j - \beta_j) + \sum_{k=1}^{d} \lambda_{jk} |\beta_{jk}| \right\}, \tag{3.1}$$

where the tuning parameters are assumed to be selected according to BIC in (2.8). Note that (3.1) defines one iteration step but with a fixed $\bar{\alpha}_j$ value. For example, for the initial step, $\bar{\alpha}_j$ is usually set to be the jth principal component (i.e., $\tilde{\beta}_j$), and then iteratively updated according to (3.1). An examination of this process can be helpful in understanding the asymptotic behavior of the fully iterated estimator. For convenience, we use $\mathcal{A}_j = \{1 \leq k \leq d : \beta_{jk} \neq 0\}$ to denote the set of nonzero coefficients in β_j and use $\hat{\mathcal{A}}_j^{\mathrm{BIC}} = \{1 \leq k \leq d : \bar{\beta}_{\lambda jk} \neq 0\}$ to denote the set of nonzero coefficients identified by $\bar{\beta}_{\lambda j}$, where the tuning parameters are selected by the BIC criterion (2.8). We have the following theorem:

Theorem 1. *Assume that $\bar{\alpha}_j - \beta_j = O_p(n^{-1/2})$ and that $\tilde{\Omega}$ converges in probability to some positive definite matrix Ω, $\sqrt{n}a_n \to 0$, and $\sqrt{n}b_n \to \infty$. We have* (1) $\bar{\beta}_{\lambda j} - \beta_j = O_p(n^{-1/2})$ *and* (2) $P(\bar{\beta}_{\lambda jk} = 0) \to 1$ *for every* $\beta_{jk} = 0$.

Theorem 2. *Assume that $\bar{\alpha}_j - \beta_j = O_p(n^{-1/2})$ and that $\tilde{\Omega}$ converges in probability to some positive definite matrix Ω. We have $P(\hat{\mathcal{A}}_j^{\mathrm{BIC}} = \mathcal{A}_j) \to 1$.*

The proofs of these two theorems follow arguments similar to those in Wang and Leng (2007). Thus, the details are omitted. By Theorem 1, we know that as long as $\bar{\alpha}_j$ is $\sqrt{n}$-consistent, $\sqrt{n}a_n \to 0$ and $\sqrt{n}b_n \to 0$, the resulting estimator $\bar{\beta}_{\lambda j}$ is $\sqrt{n}$-consistent, and all sparse loadings can be identified consistently. Theorem 2 says that as long as the BIC criterion (2.8) is used in either local or global form, the true model is guaranteed to be identified consistently.

3.2 THE OPTIMAL CHOICE OF THE KERNEL MATRIX

By Theorems 1 and 2, it seems that the kernel matrix Ω does not matter asymptotically as long as it converges to a positive definite matrix. Although Wang and Leng (2007) provided some motivation to use $\Omega = \mathrm{cov}^{-1}(\tilde{\beta}_j)$, this choice is not theoretically justified. To gain some theoretical insight, we compare the asymptotic efficiency of $\bar{\beta}_{\lambda j}$ by fixing

$\bar{\alpha}_{\lambda j} = \tilde{\beta}_j$ (i.e., the one-step estimator). In what follows, we shall show that the one-step estimator with $\Omega = \mathrm{cov}^{-1}(\tilde{\beta})$ has the "smallest" asymptotic covariance. If the one-step estimator with $\Omega = \mathrm{cov}^{-1}(\tilde{\beta})$ is asymptotically efficient, it is reasonable to expect that its fully iterated version performs better than the estimators defined by other kernel matrices.

First, according to whether the true loading coefficients are 0 or not, we partition the β_j into two parts, that is, $\beta_j = (\beta_{j,a}^\top, \beta_{j,b}^\top)^\top$ with $\beta_{j,a} = \{\beta_{jk} : \beta_{jk} \neq 0\}$ and $\beta_{j,b} = \{\beta_{jk} : \beta_{jk} = 0\}$. Their counterparts for other estimators are similarly defined. We partition the kernel matrix Ω and $C_j = \mathrm{cov}(\tilde{\beta}_j)$ accordingly as

$$\Omega = \begin{bmatrix} \Omega_{aa} & \Omega_{ab} \\ \Omega_{ba} & \Omega_{bb} \end{bmatrix} \qquad \text{and} \qquad C_j = \begin{bmatrix} C_{j,aa} & C_{j,ab} \\ C_{j,ba} & C_{j,bb} \end{bmatrix}.$$

Recall that we use the notation $\hat{\beta}_{\lambda j}^*$ to denote the estimator obtained by minimizing (2.7) with $\tilde{\Omega} = \hat{C}_j^{-1}$ (i.e., the GAS-PCA estimator). We use $\hat{\beta}_{\lambda,j}^{\tilde{\Omega}}$ to denote the estimator obtained by (2.7) but with a general matrix $\tilde{\Omega}$. Under the condition $\sqrt{n}a_n \to 0$ and $\sqrt{n}b_n \to 0$, we know by Theorem 2 that $P(\hat{\beta}_{\lambda j,b}^{\tilde{\Omega}} = 0) \to 1$. By Wang and Leng (2007, equations (A5) and (A6), p. 1047), we know that

$$\hat{\beta}_{\lambda j,a}^{\tilde{\Omega}} - \beta_{j,a} = (\tilde{\beta}_{j,a} - \beta_{j,a}) + \Omega_{aa}^{-1}\Omega_{ab}\tilde{\beta}_{j,b} + o_p\big(n^{-1/2}\big). \tag{3.2}$$

On the other hand, if we fix $\tilde{\Omega} = \hat{C}_j^{-1}$, Wang and Leng (2007) have verified that

$$\hat{\beta}_{\lambda j,a}^{*} - \beta_{j,a} = (\tilde{\beta}_{j,a} - \beta_{j,a}) - C_{j,ab}C_{j,bb}^{-1}\tilde{\beta}_{j,b} + o_p\big(n^{-1/2}\big). \tag{3.3}$$

Next, simple algebra gives

$$\begin{aligned} \mathrm{cov}(\hat{\beta}_{\lambda j,a}^{\tilde{\Omega}} - \beta_{j,a}) = {} & \mathrm{cov}(\hat{\beta}_{\lambda j,a}^{*} - \beta_{j,a}) + \mathrm{cov}(\hat{\beta}_{\lambda j,a}^{*} - \hat{\beta}_{\lambda j,a}^{\tilde{\Omega}}) \\ & - \mathrm{cov}(\hat{\beta}_{\lambda j,a}^{*} - \beta_{j,a}, \hat{\beta}_{\lambda j,a}^{*} - \hat{\beta}_{\lambda j,a}^{\tilde{\Omega}}) \\ & - \mathrm{cov}(\hat{\beta}_{\lambda j,a}^{*} - \hat{\beta}_{\lambda j,a}^{\tilde{\Omega}}, \hat{\beta}_{\lambda j,a}^{*} - \beta_{j,a}). \end{aligned} \tag{3.4}$$

Because $\hat{\beta}_{\lambda j,a}^{*} - \hat{\beta}_{\lambda j,a}^{\tilde{\Omega}} = O_p(n^{-1/2})$ and also by (3.3), we know that

$$\begin{aligned} & -\mathrm{cov}(\hat{\beta}_{\lambda j,a}^{*} - \beta_{j,a}, \hat{\beta}_{\lambda j,a}^{*} - \hat{\beta}_{\lambda j,a}^{\tilde{\Omega}}) \\ & \quad = -\mathrm{cov}((\tilde{\beta}_{j,a} - \beta_{j,a}) - C_{j,ab}C_{j,bb}^{-1}\tilde{\beta}_b, \hat{\beta}_{\lambda j,a}^{*} - \hat{\beta}_{\lambda j,a}^{\tilde{\Omega}}) + o_p(n^{-1}). \end{aligned}$$

Similarly by (3.2)–(3.3), we have

$$\begin{aligned} & -\mathrm{cov}((\tilde{\beta}_{j,a} - \beta_{j,a}) - C_{j,ab}C_{j,bb}^{-1}\tilde{\beta}_b, \hat{\beta}_{\lambda j,a}^{*} - \hat{\beta}_{\lambda j,a}^{\tilde{\Omega}}) + o_p(n^{-1}) \\ & \quad = \mathrm{cov}\big((\tilde{\beta}_{j,a} - \beta_{j,a}) - C_{j,ab}C_{j,bb}^{-1}\tilde{\beta}_b, (C_{j,ab}C_{j,bb}^{-1} + \Omega_{aa}^{-1}\Omega_{ab})\tilde{\beta}_b\big) + o_p(n^{-1}) \\ & \quad = \big(\mathrm{cov}(\tilde{\beta}_{j,a} - \beta_{j,a}, \tilde{\beta}_b) - C_{j,ab}C_{j,bb}^{-1}\,\mathrm{cov}(\tilde{\beta}_{j,b}, \tilde{\beta}_{j,b})\big) \\ & \qquad \times (C_{j,ab}C_{j,bb}^{-1} + \Omega_{aa}^{-1}\Omega_{ab})^\top + o_p(n^{-1}) \\ & \quad = (C_{j,ab} - C_{j,ab}C_{j,bb}^{-1}C_{j,bb})(C_{j,ab}C_{j,bb}^{-1} + \Omega_{aa}^{-1}\Omega_{ab})^\top + o_p(n^{-1}) = o_p(n^{-1}). \end{aligned}$$

The fourth term in (3.4) is $o_p(n^{-1})$. Thus, the following theorem holds immediately.

Theorem 3. *Assume that $\tilde{\Omega}$ converges in probability to some positive definite matrix Ω, $\sqrt{n}a_n \to 0$, and $\sqrt{n}b_n \to \infty$. We have* $\operatorname{cov}(\hat{\beta}^{\tilde{\Omega}}_{\lambda j,a} - \beta_{j,a}) = \operatorname{cov}(\hat{\beta}^{*}_{\lambda j,a} - \beta_{j,a}) + \operatorname{cov}(\hat{\beta}^{*}_{\lambda j,a} - \hat{\beta}^{\tilde{\Omega}}_{\lambda j,a}) + o_p(n^{-1})$.

The matrix $\operatorname{cov}(\hat{\beta}^{*}_{\lambda j,a} - \hat{\beta}^{\tilde{\Omega}}_{\lambda j,a})$ is nonnegative definite. Thus, Theorem 3 implies that the optimal asymptotic efficiency of $\hat{\beta}_{\lambda j}$ is achieved when $\tilde{\Omega} = \tilde{C}_j^{-1}$. This justifies the use of $\Omega = \operatorname{cov}^{-1}(\tilde{\beta}_j) = C_j^{-1}$ as proposed in Section 2.3. Furthermore, by (3.3), one can verify that

$$\operatorname{cov}(\hat{\beta}^{*}_{\lambda,a} - \beta_{j,a}) = C_{j,aa} - C_{j,ab}C^{-1}_{j,bb}C_{j,ba} + o_p(n^{-1}). \tag{3.5}$$

Because the matrix $C_{j,ab}C^{-1}_{j,bb}C_{j,ba}$ is nonnegative definite, (3.5) implies that the estimator in (2.7) with the optimal kernel matrix estimates the nonzero coefficients more efficiently than the unpenalized estimator.

4. NUMERICAL STUDIES

4.1 SIMULATION STUDIES

To compare the finite sample performance of PCA, S-PCA, SAS-PCA, GAS-PCA, and VAST, simulation studies are conducted and reported in this subsection. For each simulation example, we record the average of the estimated model size (i.e., number of nonzero loadings), the frequency of the models that are correctly identified (i.e., the situation with zero and nonzero loadings correctly identified), and the median of the average angle between the estimates and the true loadings in degrees, defined as $d_0^{-1}\sum_{j=1}^{d_0} 180\cos^{-1}(|\hat{\beta}_j^{\top}\beta_j|)/\pi$, whenever applicable. Various sample sizes ($n = 50$, 100, 200, 400, 800, 1600) are evaluated. For each model and sample size, a total of 100 simulation replications are conducted.

Example 1: For this example, we generate normal variates independently from $N(0, \Sigma)$, where Σ is specified according to the following three cases:

(A) We take $d = 8$ with $d_0 = 4$ and

$$\Sigma = \sum_{j=1}^{8} \delta_j \beta_j \beta_j^{\top},$$

where $\delta^{\top} = (4, 3, 2, 1, 0.6, 0.5, 0.4, 0.3)$, and

$$(\beta_1, \ldots, \beta_8) = \begin{pmatrix} 0.5 & 0 & 0.5 & 0 & 0.5 & 0.5 & 0 & 0 \\ 0.5 & 0 & -0.5 & 0 & 0.5 & -0.5 & 0 & 0 \\ 0.5 & 0 & 0.5 & 0 & -0.5 & -0.5 & 0 & 0 \\ 0.5 & 0 & -0.5 & 0 & -0.5 & 0.5 & 0 & 0 \\ 0 & 0.5 & 0 & 0.5 & 0 & 0 & 0.5 & 0.5 \\ 0 & 0.5 & 0 & -0.5 & 0 & 0 & 0.5 & -0.5 \\ 0 & -0.5 & 0 & -0.5 & 0 & 0 & 0.5 & 0.5 \\ 0 & -0.5 & 0 & 0.5 & 0 & 0 & 0.5 & -0.5 \end{pmatrix}.$$

(B) We take $d = 10$ with $d_0 = 2$. The eigenvalues of Σ are taken to be $(200, 100, 50, 25, 10, 5, 4, 3, 2, 1)^\top$, and the two leading eigenvectors are given by $\beta_1 \propto (1, 1, 1, 1, 0, 0, 0, 0, 0.9, 0.9)^\top$ and $\beta_2 \propto (0, 0, 0, 0, 1, 1, 1, 1, -0.3, 0.3)^\top$, respectively (Shen and Huang 2007). To generate the other eigenvectors, we draw independent and identically distributed uniform random numbers to construct a 10×8 matrix and then use the Gram–Schmidt method to orthogonalize the vectors with respect to β_1 and β_2.

(C) We generate a 12-dimensional random normal vector from $N(0, \Sigma)$ with $d_0 = 3$, where the eigenvalues of Σ are $(100, 60, 30, 10, 10, 10, 5, 5, 5, 5, 2, 2)^\top$, and the three leading eigenvectors are given by

$$\beta_1 \propto (1, 1, 0.2, 0.2, 0, 0, 0, 0, 0, 0, 0, 0),$$

$$\beta_2 \propto (0, 0, 0, 0, 1, 1, 0.1, 0.1, 0, 0, 0, 0),$$

$$\beta_3 \propto (0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 0.3, 0.2).$$

The other eigenvectors are generated according to case (B).

The loadings in cases (B) and (C) are not well separated. The results are summarized in Figure 1, from which several observations can be made. First, SAS-PCA generally outperforms S-PCA in variable selection, although their estimation accuracies may be comparable. Second, in terms of variable selection, VAST may perform better (case (C)) or worse (case (A)) than SAS-PCA but always worse than GAS-PCA; see Panels (b), (e), and (h) in Figure 1. Overall, GAS-PCA performs best in terms of estimation accuracy and variable selection. These observations confirm our asymptotic results.

Example 2: To shed further light on the estimators we wish to compare, we take an oracle sparsity approach by fitting the model with the true sparsity level. Thus, we assume that, for each principal component we know exactly the true sparsity level (i.e., the number of zero entries in a loading), but do not know where the zero loadings are located. Because the true sparsity level is known, we can search for these estimates which satisfy that sparsity requirement perfectly on the solution path. The final estimate corresponds to the smallest tuning parameter satisfying this requirement. The objective of such an experiment is to compare the aforementioned estimation methods, if one can indeed deduce the "correct" number of nonzeros as in Zou, Hastie, and Tibshirani (2006). To save space, only case (C) in Example 1 is reported, as the findings for other cases are qualitatively similar. For this example, the oracle sparsity is $(8,\ 8,\ 8)$ for the three leading eigenvectors. Because the model size is always 12, we recorded instead the average number of zero coefficients that are correctly/incorrectly identified. Results presented in Figure 2 show that GAS-PCA performs similarly to VAST in variable selection, but gives better results in estimation accuracy. The performances of other methods (i.e., PCA, S-PCA, and SAS-PCA) are much worse.

Example 3: This example is used to evaluate the effect of one-step versus multi-step estimation and also the local versus global BIC methods for GAS-PCA. All possible combinations result in four estimators and we shall denote them respectively by GBIC-m,

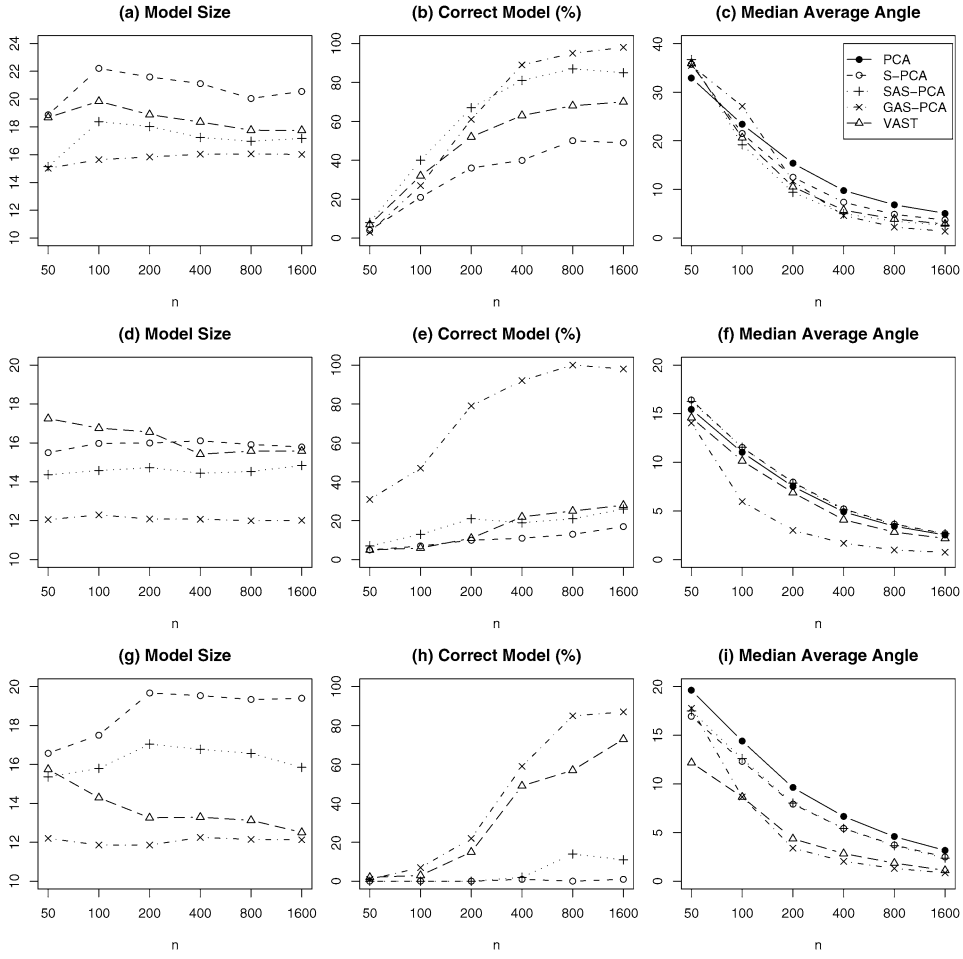


Figure 1. Simulation comparing PCA, S-PCA, SAS-PCA, GAS-PCA, and VAST in Example 1. (a)–(c) For case (A); (d)–(f) for case (B); (g)–(i) for case (C).

GBIC-1, LBIC-m, and L-BIC-1, where G refers to "global," L refers to "local," 1 to "one-step," and m to "multistep." To save space, only the results of case (C) from Example 1 are reported. The results summarized in Figure 3 clearly suggest that the multiple-step GAS-PCA estimator with the local BIC method has the best finite sample performance. The other two cases in Example 1 also produce similar findings.

4.2 THE TEACHING EVALUATION DATA

We illustrate the usefulness of the proposed methods via an example of teaching evaluation data. This dataset is about the teaching evaluation scores of 251 courses taught in the Guanghua School of Management, Peking University. Each observation corresponds to one course taught during the period from 2002 to 2004, and records the average scores on the students' agreement with the following nine statements: (Q1) I think this is

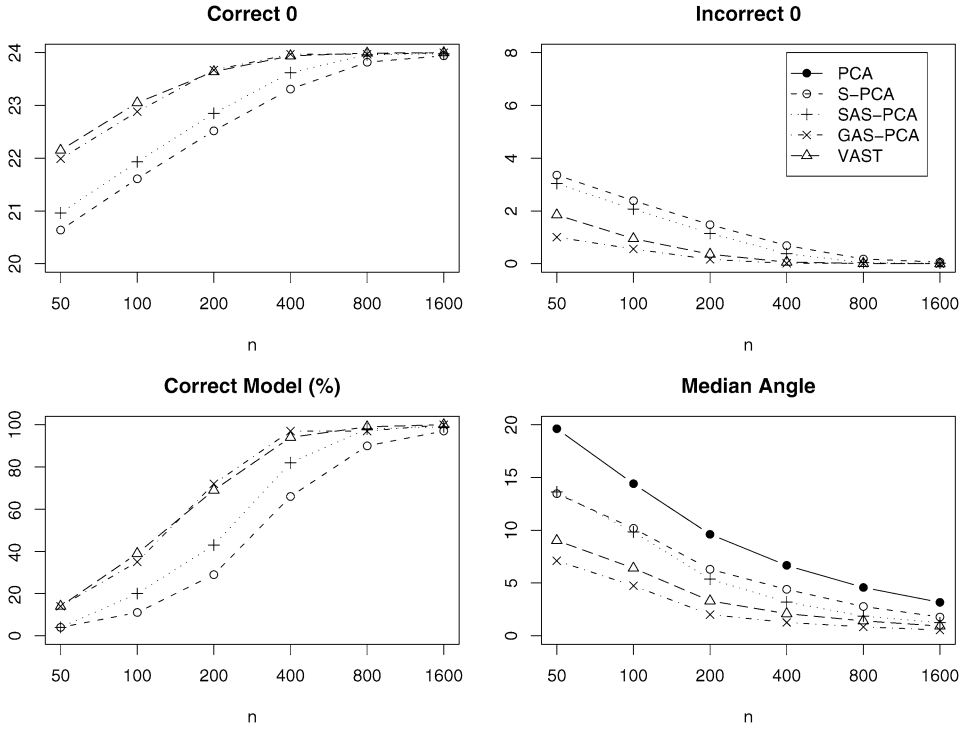


Figure 2. Simulation comparing PCA, S-PCA, SAS-PCA, GAS-PCA, and VAST in Example 2 using the oracle sparsity.

a good course; (Q2) The course improves my knowledge; (Q3) The schedule is reasonable; (Q4) The course is difficult; (Q5) The course pace is too fast; (Q6) The course load is very heavy; (Q7) The text book is good; (Q8) The reference book is helpful; (Q9) Opening this course is necessary. Various methods (i.e., PCA, S-PCA, SAS-PCA, GAS-PCA, and VAST) are applied to this dataset. Starting from the fourth principal component, no

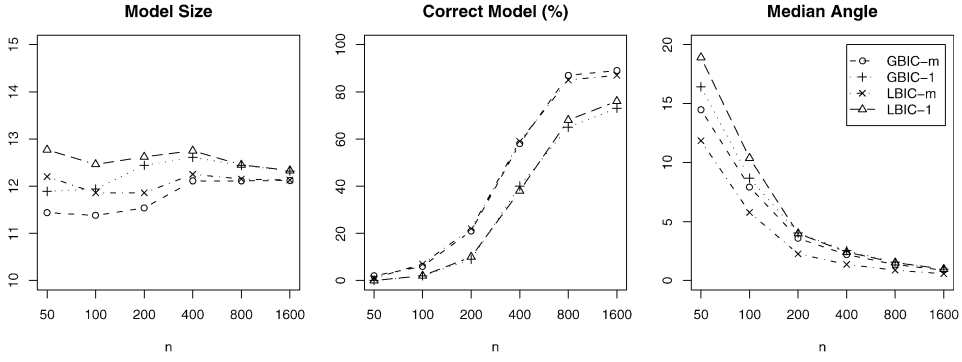


Figure 3. Simulation comparing local BIC and global BIC, and multiple-steps estimator and one-step estimators for GAS-PCA. LBIC-m: local BIC with multiple-step iterations; LBIC-1: local BIC with one-step iteration; GBIC-m: global BIC with multiple-step iterations; GBIC-1: global BIC with one-step iteration.

Table 1. The teaching evaluation data analysis results.

	S-PCA			SAS-PCA			GAS-PCA			VAST		
	PC1	PC2	PC3	PC1	PC2	PC3	PC1	PC2	PC3	PC1	PC2	PC3
Q1	0.49		0.32	0.61		0.44	0.52		0.31	0.47		0.32
Q2	0.35		0.34			0.41	0.39		0.32	0.37		0.34
Q3	0.35		0.31			0.40	0.29		0.22	0.34		0.29
Q4		0.62			0.60			0.64		0.08	0.63	
Q5		0.56			0.58			0.51		−0.05	0.53	
Q6		0.55			0.56			0.57		−0.03	0.57	
Q7	0.50		−0.64	0.64		−0.58	0.42		−0.67	0.48		−0.64
Q8	0.40		−0.43	0.47			0.39		−0.48	0.41		−0.45
Q9	0.33		0.31			0.38	0.41		0.28	0.36		0.31

method is able to deliver easily interpretable results. Thus, we focus on the first three principal components (i.e., $d_0 = 3$), which account for about 82% of total variability according to the PCA analysis.

The detailed estimation results are given in Table 1, from which we find that the loading patterns identified by both SAS-PCA and VAST can hardly be interpreted. On the other hand, both S-PCA and GAS-PCA identified similar loading patterns, which can be easily interpreted. Specifically, PC2 clearly suggests that the items Q4, Q5, and Q6 are highly correlated with each other. A closer look of the questionnaire reveals that those three items are the only questions asked in a negative manner. For example, instead of asking students whether they agree that the course is easy, item Q4 asks students whether they agree that the course is difficult. Similar comments are applicable to Q5 and Q6. This pattern provides practitioners with solid evidence that the way the questions are asked does affect the responses. On the other hand, we find that Q7 and Q8 are the only two items having negative loadings in PC3. Furthermore, the loading patterns of other items in PC3 are similar to those of PC1 for S-PCA and GAS-PCA. The results suggest that Q1, Q2, Q3, and Q9 are likely to be accounted for by one common factor whereas Q7 and Q8 are due to some other factors. Recalling the meaning of those items, we find that Q7 and Q8 are the only two items about teaching materials (i.e., textbook or references), whereas all other items (i.e., Q1, Q2, Q3, and Q9) are general questions about the course itself.

Although similar loading patterns are identified by S-PCA and GAS-PCA, their estimated loading coefficients are different. Our simulation experiments presented in the previous subsection suggest that GAS-PCA typically gives a more accurate estimate than S-PCA. However, for a real dataset without the "true parameter," it is difficult to assess which estimate is more accurate. To partially resolve this problem, we conduct an experiment in favor of S-PCA. Namely, we take the S-PCA estimate as the true parameter and then simulate 251 observations in a manner similar to Example 1. Thus, we know the true loading patterns of the first three principal components, which are given by the S-PCA estimate in Table 1. We then compare the estimation accuracy of S-PCA and GAS-PCA in terms of the median average angle as defined in the previous section. We find that, if the oracle sparsity approach is used for both methods, the median average angles of S-PCA

and GAS-PCA are given by 5.04 and 4.07, respectively. On the other hand, if BIC are used for selecting the tuning parameters, the corresponding median average angles are 8.48 and 4.10, respectively. In either case, it seems that the GAS-PCA estimate may be more accurate than the S-PCA estimate. Similar experiments are conducted for the other three methods (i.e., PCA, SAS-PCA, and VAST). We observe the same phenomenon.

5. CONCLUDING REMARKS

In this article we investigate a few methods for improving S-PCA in terms of adaptiveness and efficiency. The adaptiveness is achieved by replacing the lasso penalty by the adaptive lasso penalty, whereas higher efficiency is obtained by formulating a more general least-squares objective function with an optimal choice of kernel matrix. The latter effort has a clear connection to the least-squares approximation method of Wang and Leng (2007), and thus its theoretical properties can be similarly established as in Wang and Leng (2007). As a by-product, our general formulation also allows us to investigate soft-thresholding types of estimators. How to extend GAS-PCA and other approaches in the article to other multivariate analyses is an interesting topic for future research.

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